MIL-T-16644D
30 September 1986
SUPERSEDING
MIL-T-16644C
30 April 1970

MILITARY SPECIFICATION

TAPES, MEASURING: TANK GAGING TYPE, WITH REEL AND PLUMB BOB

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers reel-mounted, steel measuring tapes, with plumb bob, for gaging the level of liquids in tanks.

1.2 Classification. The measuring tapes shall be of the following lengths as specified (see 6.2):

- 25 feet
- 50 feet
- 75 feet
- 100 feet

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5210

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-18T09:16Z
Check the source to verify that this is the current version before use.

SPECIFICATION

FEDERAL

PPP-P-40 - Packaging and Packing of Hand Tools

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard product. The tapes covered by this document shall be the regular commercial product of the manufacturer except for changes necessary to comply with document requirements.

* 3.2 Materials. Materials not definitely specified shall be of the quality normally used by the contractor for reel-mounted measuring tapes, with plumb bobs, provided that the completed item complies with all provisions of this document. Recycled material may be used (see 6.3).

3.3 Construction. The reel-mounted measuring tape, with plumb bob, shall consist of an assembly of component parts as shown in figure 1. When uncoiled, the tape shall be removable from the reel and the reel from the frame without the need for any tools. The assembly shall be capable of withstanding the load test specified in 4.3.1 without breaking or permanent set and shall work freely by the weight of the plumb bob after the test.

* 3.3.1 Frame. The frame material shall be carbon steel treated to resist corrosion by electroplating or corrosion-resisting steel. Painting shall not be used as a means of corrosion resistance or protection. The frame shall consist of two sides held apart by spacers permanently secured in place.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:16Z
Check the source to verify that this is the current version before use.

3.3.2 Handle. The handle shall be made of hardwood that is enameled or polished and shall be of a size to ensure a firm hold. The handle shall be fastened to the frame by an axial rod in such a manner as to insure a firm and secure assembly.

* 3.3.3 Tape wiper. The tape wiper shall be plated steel, brass, bronze, or aluminum and shall include a device for adjusting the tension of replaceable felt wiper pads. When the tape is reeled in and locked in place, the tape wiper shall have provisions for supporting the extreme end of the tape so as to prevent bending of the tape from a swinging plumb bob.

* 3.3.4 Reel. The reel shall be not less than 2 inches nor greater than 3 3/4 inches in diameter.

3.3.5 Winding arm. The winding arm shall be fitted with a finger knob that shall revolve independently of the arm.

3.3.6 Spindle. The spindle shall be constructed so that it rotates freely in the frame when the winding arm is turned and causes the reel to rotate without binding or coming in contact with the frame.

* 3.3.7 Tape. The tape shall be spring-tempered steel not less than 1/2-inch wide and not less than 0.008-inch nor greater than 0.013-inch thick. The tape shall be fitted with a swivel snap at the free end to accommodate a plumb bob. The tape shall withstand coiling around a 1 3/4-inch rod without permanent set (see 4.3.2). The end of the tape shall be reinforced (see figure 1) to minimize bending of the tape at the extreme end.

3.3.7.1 Graduation. Graduation on the tape shall indicate feet, inches, and eighths of inches on one side only, with the first foot including the plumb bob and swivel snap. The contact point in the swivel snap (for plumb bob, inside of lower end of snap) shall represent 6.600 inches $\pm$ 0.016 inch. The error, when including plumb bob and swivel snap from zero to the 1-foot graduation, shall be not greater than $\pm$ 1/32 inch. The error in the total length of the tape shall not exceed 0.100 inch in 100 feet, when the tape is supported on a horizontal surface at 68°F and subjected to a tension load of 10 pounds (see 4.3.3). The graduation lines shall be not more than 1/32-inch wide. Each foot and inch increment shall be numbered and shall be distinctive in character.

* 3.3.7.2 Tape finish. The tape shall be supplied with raised or recessed numerals and graduations in natural steel color, raised or recessed not less than 0.0005 inch, clearly legible on a black etched background, or black graduations and markings on a light rust-resistant background, as specified (see 6.1 and 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:16Z
Check the source to verify that this is the current version before use.

3.3.7.3 Tape length. The length of graduated tape with swivel snap shall be not less than 25, 50, 75, or 100 feet, as specified (see 6.2).

3.3.7.4 Tape refills. When specified (see 6.2), refill tapes, with swivel snaps only, shall be furnished. The refill tapes shall be of the length specified in the contract or purchase order and shall conform to 3.3.7 through 3.3.7.3.

3.3.8 Plumb bob. The plumb bob shall be of brass and shall be 1 inch at the largest diameter and approximately 6 3/4 inches long. It shall have a section of the surface approximately 3/16 inch wide, full length of the 1-inch diameter, serrated to retain oil for easy reading. It shall have a tapered end with a flattened point not to exceed 5/32 inch in diameter on one end, and the other end (top) shall be furnished with a lug or ear and eye, with chamfers or radii, for the swivel snap attachment. The distance between the flattened end and contact point in eye of plumb bob (upper edge of hole) shall be 6.600 inches $\pm$ 0.016 inch. The plumb bob shall have a rated weight of 20 ounces. When specified (see 6.2), the plumb bob only shall be furnished.

3.4 Marking. The end item shall be permanently and legibly marked with the manufacturer's name or readily identifiable trademark. In addition, the frame or winding arm shall be marked with the tape length.

3.5 Workmanship. The end item shall be clean and smooth on all surfaces. The handles shall have no splits, seams, cracks, splinters, or gouges. Painted or enameled surfaces shall have no uncoated area, no chipped film, and no discoloration. All metal parts shall be free of sharp edges, fins, burrs, splits, and machine marks. Plating, when applicable, shall show no oxide scale, rust, roughness, or foreign inclusions and shall be smooth and uniform with no uncoated, nonadherent, or peeling areas.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-18T09:16Z
Check the source to verify that this is the current version before use.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of tapes. The sample unit shall be one tape. The inspection level shall be S-2, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

TABLE I. End item visual defect

Examine	Defect
Construction and workmanship	Materials not as specified
	Frame spacers not permanently secured in place
	Handle missing or damaged; evidence of splits, cracks, seams, or splintering; finish not as specified; handle not secured to frame as specified
	Tape wiper nonconforming or not as specified; tension of the felt wiper pads on tape not adjustable; wiper improperly located
	Winding arm not as specified; finger knob missing, damaged, or fails to revolve independently of the arm
	Failure of spindle to rotate freely in the frame; evidence of binding or spindle coming in contact with the frame

Source: https://assist.dla.mil -- Downloaded: 202610-18T09:16Z
Check the source to verify that this is the current version before use.

TABLE I. End item visual defect (cont'd)

Examine	Defect
Construction and workmanship (cont'd)	Tape dimensions not as specified; swivel snap missing or not attached to free end of tape; evidence of binding or friction between tape and frame, handle, or reel; tape not attached to the reel
	Graduations and figures missing, incomplete, or illegible; difference in size or fractional graduations not readily distinguishable; graduations not in feet, inches, and fraction of an inch as specified; error in graduation; each foot and inch increment not properly numbered nor distinctive in character
	Tape finish nonconforming or not as specified; raised or recessed markings and graduations less than the specified minimum
	Plumb bob shape nonconforming or not as specified; oil retaining serrations missing or not as specified
	Length of tape(s) not as specified
	Exposed metal parts not free from sharp edges, burrs, fins, roughness, or foreign inclusions
	Removal of the tape from the reel and the reel from the frame necessitates the use of tools
	Plating, when applicable, incomplete; not smooth or uniform, evidence of peeling, nonadherent or uncoated areas; evidence of oxide scale or rust
	Tape refills and replacement plumb bob, when specified, nonconforming or not as specified
	Part missing, damaged, nonconforming, or improperly fitted or assembled
Marking	Marking of manufacturer's name or trademark and the tape length missing, nonconforming, incorrect, illegible, or not permanent

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:16Z
Check the source to verify that this is the current version before use.

4.2.3 End item dimensional and weight examination. The end item shall be examined for compliance with the dimensions and plumb bob weight as specified in 3.3. Any dimensions or weight not within specified requirements shall be classified as a defect. The lot size shall be expressed in units of tapes. The sample unit shall be one tape. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.2.4 End item testing. The end item shall be tested as specified in 4.3.1 through 4.3.3. The lot size shall be expressed in units of tapes. The sample unit shall be one tape. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.2.5 Packaging examination. Examination of packaging shall be made in accordance with PPP-P-40.

4.3 Methods of inspection.

4.3.1 Load test. The assembled tape and reel shall be tested in a locked position, with not less than 1 foot of tape extended and with a 20-pound load hanging from the swivel snap. The 20-pound load shall then be removed and the tape allowed to uncoil full length, by weight of the plumb bob. Any evidence of breakage, permanent distortion, binding, or failure of the locking mechanism shall constitute failure of the test (see 3.3).

4.3.2 Coiling test. The tape shall be rolled around a rod having a nominal diameter of 1 3/4 inches and, when removed, no permanent set shall be visible (see 3.3.7). Any evidence of permanent set, when uncoiled, shall constitute failure of this test.

4.3.3 Accuracy test. The tape shall be tested in an ambient temperature of 68° ± 2°F. The uncoiled tape shall be supported on a horizontal surface, and a 10-pound tension load applied to the swivel snap. The accuracy shall be checked against a master rule, instruments, or laboratory equipment of known accuracy. Any error greater than 0.100 inch in 100 feet, throughout the length of the tape, shall constitute failure of the test (see 3.3.7.1).

5. PACKAGING

5.1 Packaging. Preservation, packing, and marking for level A, B, or Commercial shall be in accordance with the applicable requirements of PPP-P-40 (see 6.2).

6. NOTES

6.1 Intended use. Gaging tape is used for gaging the level of liquids in tanks in various services, including, but not limited to, fresh water, sea water, and oils, including fuel and lubricating oil. A dark background tape is recommended for measuring colorless or light colored liquids. A light background is recommended for measuring dark colored liquids.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:16Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Length of tape required (see 1.2 and 3.3.7.3).
c. Tape finish required (see 3.3.7.2).
d. When refill tapes, with swivel snaps only, are required (see 3.3.7.4).
e. When replacement plumb bob only is required (see 3.3.8).
f. Selection of applicable levels of preservation and packing (see 5.1).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

* 6.4 Subject term (key word) listing.

Gaging tape
Measuring instruments
Tapes

6.5 Changes from previous issue. The margins of this document have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in the notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

Army - GL

Review activities:

Army - ME, AV
Navy - MC

User activity:

Air Force - 84

Preparing activity:

Army - GL

Project no. 5210-0254

Source: https://assist.dla.mil -- Downloaded: 202610-18T09:16Z
Check the source to verify that this is the current version before use.

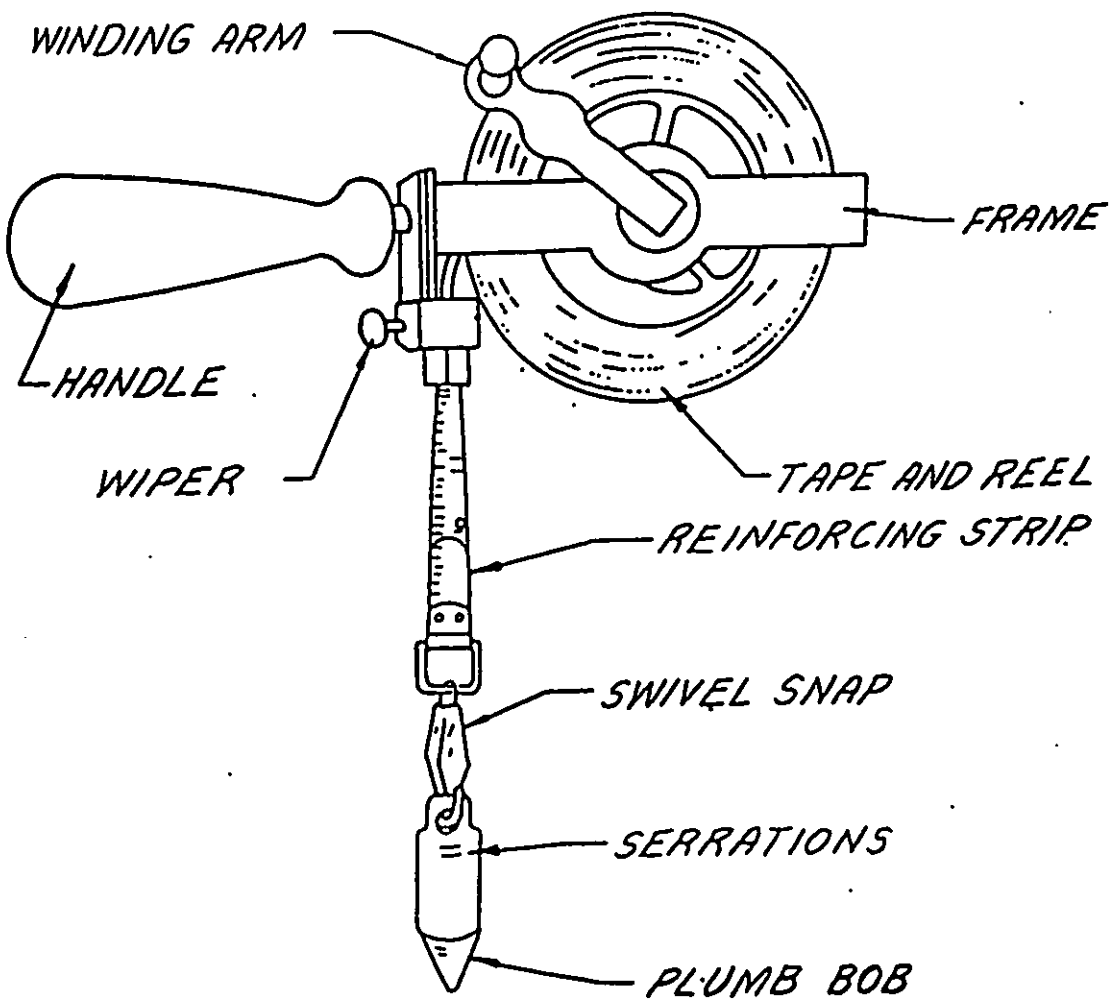


Figure 1. Tank Gaging Type Measuring Tape with Reel and Plumb Bob

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:16Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-16644D	Tapes, Measuring, Tank Gaging Type, with Reel and Plumb Bob

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2020416-181109.1oz
Check the source to verify that this is the current version before use.